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CO2_AT2 — Visualization Development Exercise
Topics: Directory of Visualizations · Visualizing Amounts · Visualizing Distributions

QUESTION-1

1. Comparing the Population of 8 Countries in the Current Year

Chart Type: Bar Chart (Amounts)

Justification:
A bar chart is the most appropriate choice for comparing the population of several countries. The length or height of the bars allows viewers to quickly compare values and identify the countries with the highest and lowest populations.

2. Showing How Delivery Times Vary Across 5 Courier Companies

Chart Type: Box Plot (Distribution)

Justification:
A box plot summarizes the distribution of delivery times by displaying the median, quartiles, overall spread, and outliers. It is useful for comparing the performance and consistency of different courier companies.

3. Showing the Proportion of a Household Budget Spent on Rent, Food, Transport, and Savings

Chart Type: Pie Chart (Proportion)

Justification:
A pie chart is effective for showing how the total household budget is divided among different expense categories. Each slice represents a portion of the total, making the contribution of each category easy to understand.

4. Showing the Relationship Between Hours Studied and Exam Marks for 50 Students

Chart Type: Scatter Plot (X–Y Relationship)

Justification:
A scatter plot is suitable for examining the relationship between study hours and exam marks. It helps reveal patterns, trends, and the strength of the correlation between the two variables.

5. Showing Average Temperature for Each Month Across 3 Different Cities

Chart Type: Line Chart (Temporal)

Justification:
A line chart clearly illustrates changes in temperature over time. Using separate lines for each city makes it easy to compare monthly temperature patterns and seasonal variations.

6. Showing Which Airports Are Directly Connected by Flight Routes

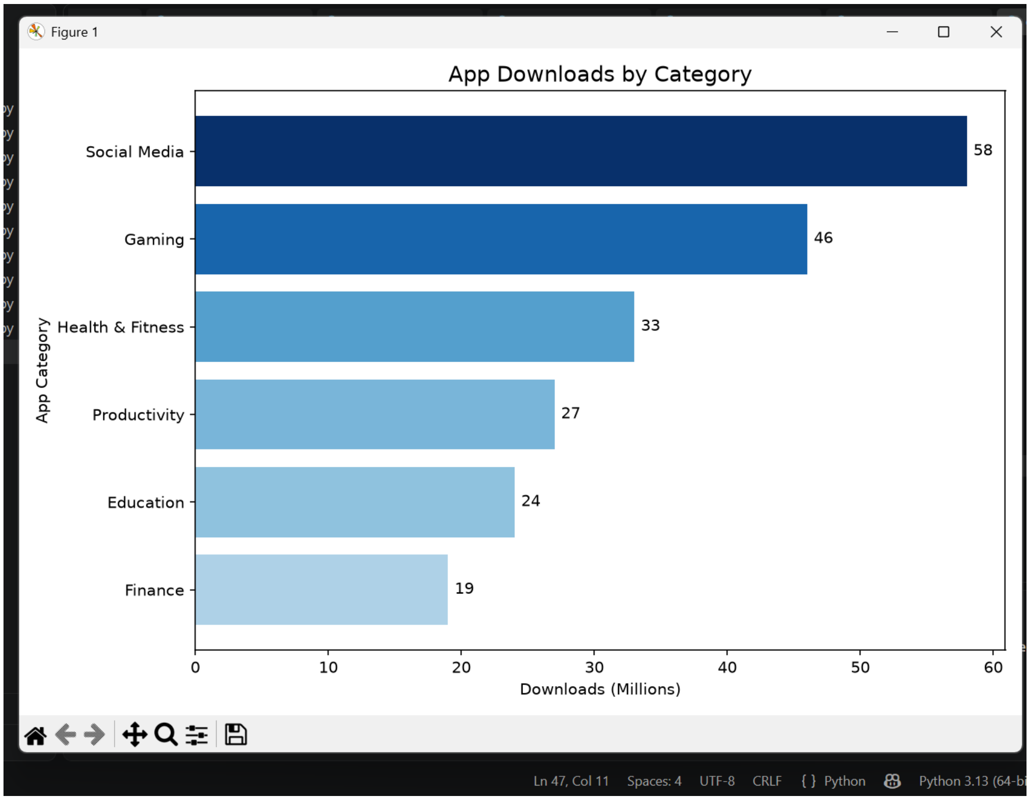
Chart Type: Network Diagram (Network)

Justification:
A network diagram is the best visualization for displaying direct connections between airports. Airports are represented as nodes, while flight routes are shown as links, making the connectivity between locations easy to interpret.

Final Summary Table

Scenario	Recommended Chart	Visualization Type
Population comparison of 8 countries	Bar Chart	Amounts
Delivery time analysis of courier companies	Box Plot	Distribution
Household budget distribution	Pie Chart	Proportion
Study hours versus exam marks	Scatter Plot	X–Y Relationship
Monthly temperature comparison of 3 cities	Line Chart	Temporal
Airport route connections	Network Diagram	Network

QUESTION-2



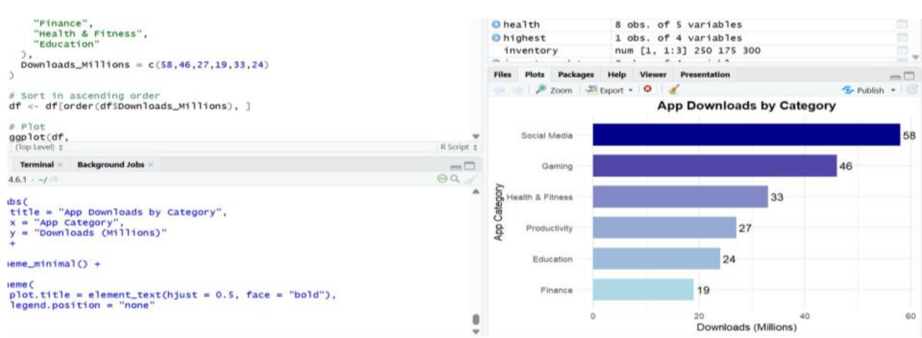
CODE:

```
import pandas as pd
import matplotlib.pyplot as plt

# Create DataFrame
df = pd.DataFrame({
    "App_Category": [
        "Social Media",
        "Gaming",
        "Productivity",
        "Finance",
        "Health & Fitness",
        "Education"
    ],
    "Downloads_Millions": [58, 46, 27, 19, 33, 24]
})

# Sort data by downloads
df = df.sort_values(by="Downloads_Millions")

# Create color gradient
```

```
colors = plt.cm.Blues(
    df["Downloads_Millions"] / df["Downloads_Millions"].max()
)

# Create horizontal bar chart
plt.figure(figsize=(9, 6))

bars = plt.barh(
    df["App_Category"],
    df["Downloads_Millions"],
    color=colors
)

# Add value labels
for bar in bars:
    plt.text(
        bar.get_width() + 0.5,
        bar.get_y() + bar.get_height() / 2,
        str(int(bar.get_width())),
        va="center",
        fontsize=10
    )

# Add title and axis labels
plt.title("App Downloads by Category", fontsize=14)
plt.xlabel("Downloads (Millions)")
plt.ylabel("App Category")

# Display chart
plt.tight_layout()
plt.show()
```

CODE:

```
# Load library
library(ggplot2)
```



```r
# Create Data Frame
df <- data.frame(
  App_Category = c(
    "Social Media",
    "Gaming",
    "Productivity",
    "Finance",
    "Health & Fitness",
    "Education"
  ),
  Downloads_Millions = c(58, 46, 27, 19, 33, 24)
)

# Sort data in ascending order
df <- df[order(df$Downloads_Millions), ]

# Create Horizontal Bar Chart
ggplot(
  df,
  aes(
    x = reorder(App_Category, Downloads_Millions),
    y = Downloads_Millions,
    fill = Downloads_Millions
  )
) +
  geom_col(width = 0.7) +
  coord_flip() +
  geom_text(
    aes(label = Downloads_Millions),
    hjust = -0.2,
    size = 4
  ) +
  scale_fill_gradient(
    low = "lightblue",
    high = "darkblue"
  ) +
  labs(
    title = "App Downloads by Category",
    x = "App Category",
    y = "Downloads (Millions)"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold"),
    legend.position = "none"
  )
```

Expected Output

The program generates a horizontal bar chart with app categories arranged in ascending order of downloads:

- Finance – 19 million
- Education – 24 million
- Productivity – 27 million
- Health & Fitness – 33 million

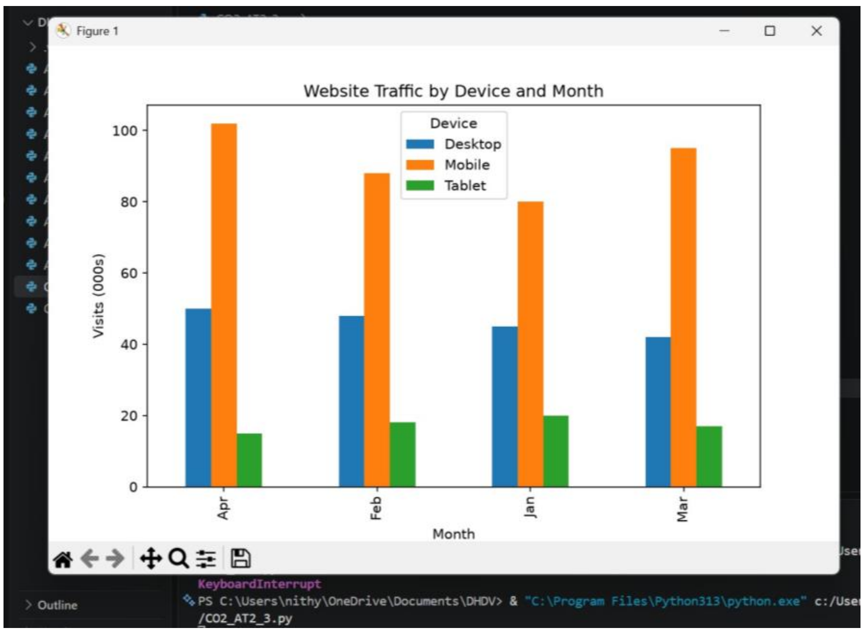
- Gaming – 46 million
- Social Media – 58 million

The bars become progressively darker as the number of downloads increases. Each bar also displays its exact download value, making the chart easy to interpret.

Interpretation

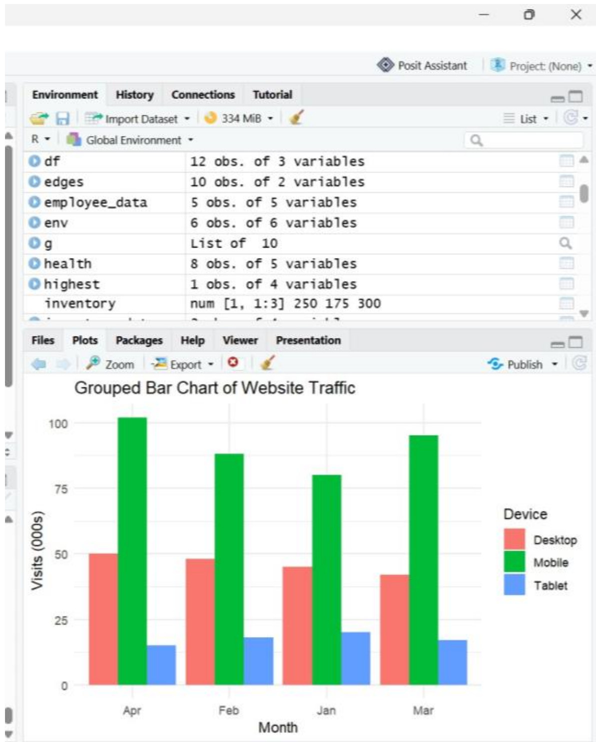
- Social Media records the highest number of downloads (58 million), making it the most popular app category.
- Gaming ranks second with 46 million downloads, indicating strong user interest.
- Health & Fitness has a moderate download count of 33 million, reflecting steady popularity.
- Productivity and Education receive 27 million and 24 million downloads, respectively, showing moderate adoption.
- Finance has the lowest number of downloads (19 million), indicating comparatively lower popularity.

QUESTION-3



CODE:

```
import pandas as pd
import matplotlib.pyplot as plt

# Create DataFrame
df = pd.DataFrame({
    "Device": [
        "Desktop", "Desktop", "Desktop", "Desktop",
        "Mobile", "Mobile", "Mobile", "Mobile",
        "Tablet", "Tablet", "Tablet", "Tablet"
    ],
    "Month": [
        "Jan", "Feb", "Mar", "Apr",
        "Jan", "Feb", "Mar", "Apr",
        "Jan", "Feb", "Mar", "Apr"
    ],
    "Visits_000s": [45, 48, 42, 50, 80, 88, 95, 102, 20, 18, 17, 15]
})

# Create Pivot Table
pivot = df.pivot(
    index="Month",
    columns="Device",
    values="Visits_000s"
)

# Plot Grouped Bar Chart
pivot.plot(kind="bar", figsize=(8, 5))

plt.title("Website Traffic by Device and Month")
plt.xlabel("Month")
plt.ylabel("Visits (000s)")
plt.legend(title="Device")

plt.tight_layout()
plt.show()
```



Code:

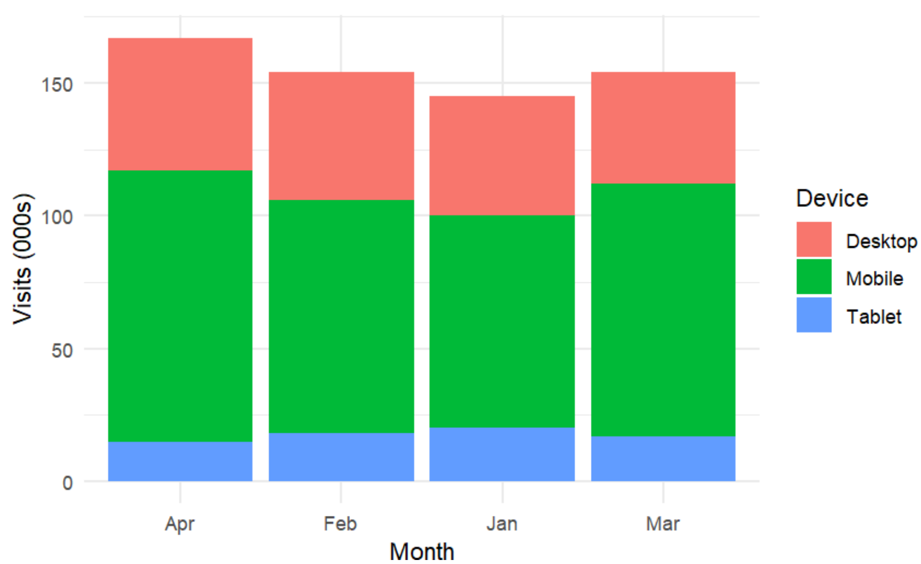
```
# Load library
library(ggplot2)

# Create Data Frame
df <- data.frame(
  Device = c(
    "Desktop", "Desktop", "Desktop", "Desktop",
    "Mobile", "Mobile", "Mobile", "Mobile",
    "Tablet", "Tablet", "Tablet", "Tablet"
  ),
  Month = c(
    "Jan", "Feb", "Mar", "Apr",
    "Jan", "Feb", "Mar", "Apr",
    "Jan", "Feb", "Mar", "Apr"
  ),
  Visits_000s = c(
    45, 48, 42, 50,
    80, 88, 95, 102,
    20, 18, 17, 15
  )
)

# Create Grouped Bar Chart
ggplot(df, aes(x = Month, y = Visits_000s, fill = Device)) +
  geom_col(position = "dodge") +
  labs(
    title = "Grouped Bar Chart of Website Traffic",
    x = "Month",
    y = "Visits (000s)"
```

```
  ) +
  theme_minimal()
```

Expected Output
The program generates a grouped bar chart displaying website traffic for Desktop, Mobile, and Tablet devices from January to April.

- Each month contains three bars representing the three device types.
- Mobile has the tallest bars across all months.
- Desktop shows moderate traffic levels.

Task 8: Stacked Bar Chart

- 

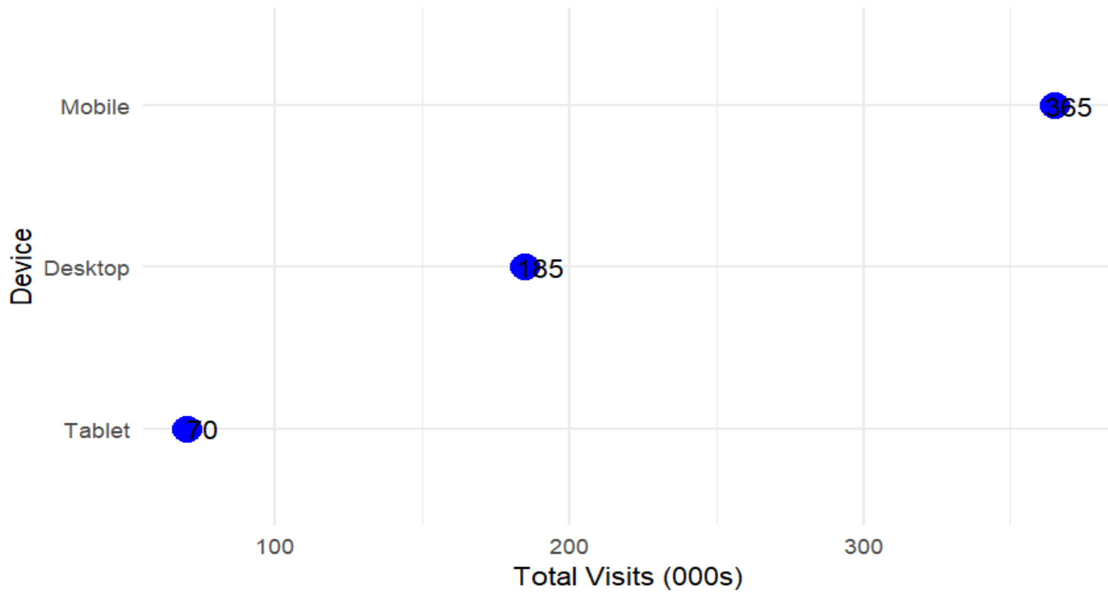
CODE:

```
# Load library
library(ggplot2)

# Create Stacked Bar Chart
ggplot(df, aes(x = Month, y = Visits_000s, fill = Device)) +
  geom_col(position = "stack") +
  labs(
    title = "Stacked Bar Chart of Website Traffic",
    x = "Month",
    y = "Visits (000s)"
  ) +
  theme_minimal()
```

Interpretation

- **April** has the highest total website traffic among all four months.
- **Mobile** devices contribute the largest share of visits every month, indicating they are the primary source of website traffic.
- **Desktop** contributes a moderate number of visits and remains fairly consistent throughout the period.

Task 9: Dot Plot



CODE:

```
# Load libraries
library(dplyr)
library(ggplot2)

# Calculate total visits for each device
total <- df %>%
  group_by(Device) %>%
  summarise(Total = sum(Visits_000s)) %>%
  arrange(desc(Total))

# Create Dot Plot
ggplot(total, aes(x = Total, y = reorder(Device, Total))) +
  geom_point(size = 5, color = "blue") +
  geom_text(
    aes(label = Total),
    nudge_x = 5
  ) +
  labs(
    title = "Dot Plot of Total Website Traffic",
    x = "Total Visits (000s)",
    y = "Device"
  ) +
  theme_minimal()
```

Interpretation

- **Mobile** records the highest total website traffic (**365 thousand visits**), indicating it is the most commonly used device for accessing the website.
- **Desktop** generates a moderate amount of traffic with **185 thousand visits**.
- **Tablet** has the lowest traffic (**70 thousand visits**), showing comparatively lower user engagement.

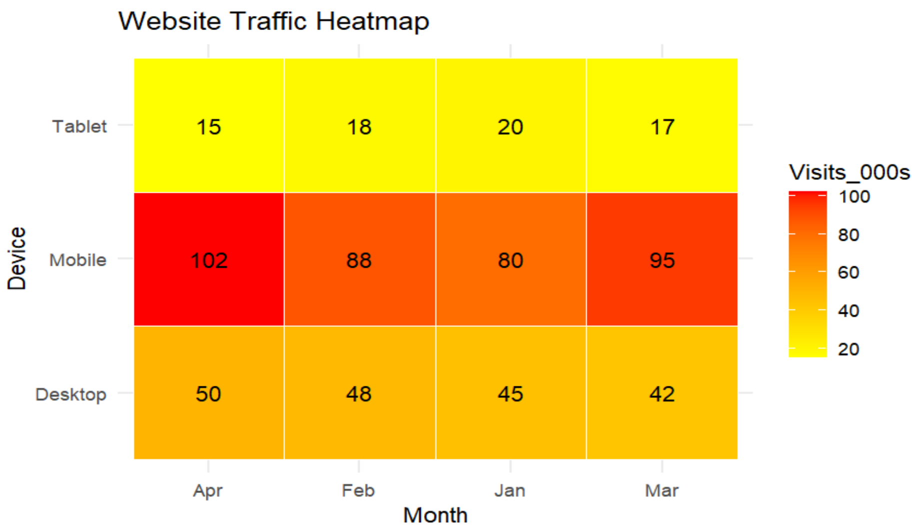
Task 10: Heatmap

```
# Load library
library(ggplot2)

# Create Heatmap
ggplot(df, aes(x = Month, y = Device, fill = Visits_000s)) +
  geom_tile(color = "white") +
  geom_text(
    aes(label = Visits_000s),
    color = "black"
  ) +
  scale_fill_gradient(
    low = "yellow",
    high = "red"
  ) +
  labs(
    title = "Website Traffic Heatmap",
    x = "Month",
    y = "Device"
  ) +
  theme_minimal()
```

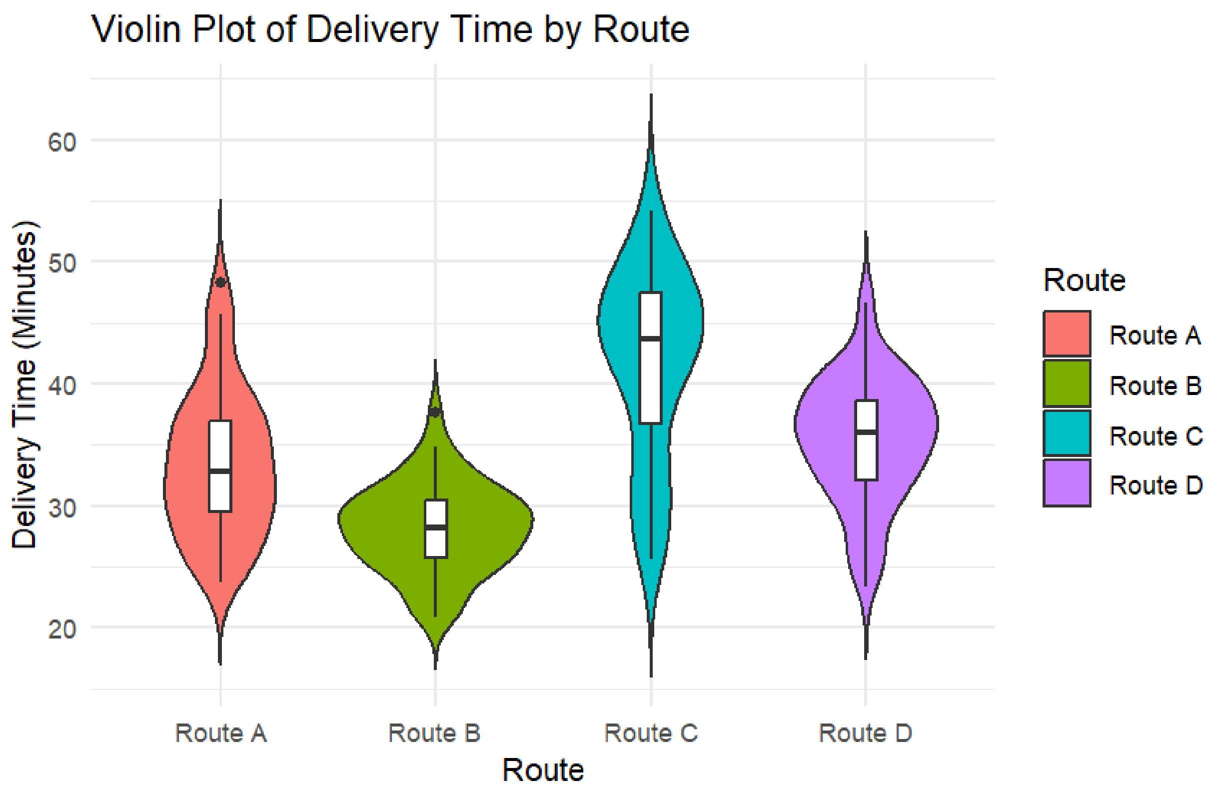
Expected Outputs

1. Grouped Bar Chart
   - Displays website traffic for Desktop, Mobile, and Tablet side by side for each month.
   - Makes it easy to compare traffic across device types within the same month.
2. Stacked Bar Chart
   - Shows the total website traffic for each month.
   - Different colored sections represent the contribution of each device category.
3. Dot Plot
   - Presents the total website visits for each device.

- Devices are ranked from the highest to the lowest total traffic.



QUESTION-4


Violin Plot of Delivery Time by Route
Delivery Time (Minutes)
60
50
40
30
20
Route A
Route B
Route C
Route D
Route
Route
Route A
Route B
Route C
Route D

CODE:

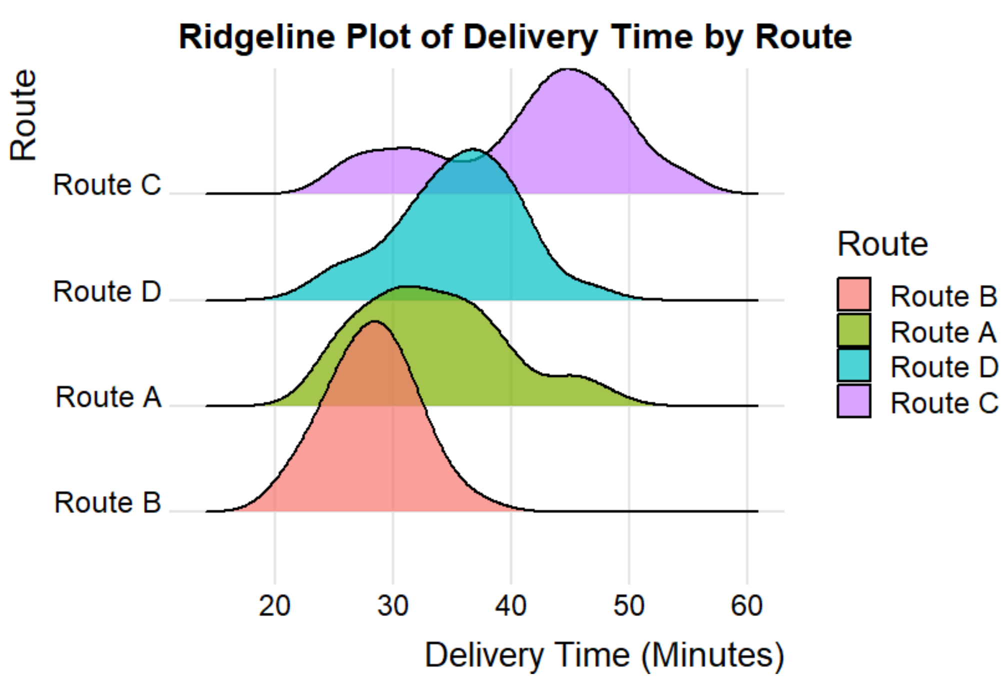
```
library(ggplot2)

# Set seed

set.seed(7)

# Generate data

routeA <- rnorm(50,32,6)

routeB <- rnorm(50,28,4)

routeC <- rnorm(50,40,9)

routeD <- rnorm(50,35,5)

# Data Frame

df <- data.frame(

Route=c(rep("Route A",50),

      rep("Route B",50),

      rep("Route C",50),

      rep("Route   D",50)),

Delivery_Time=c(routeA,

            routeB,

            routeC,

            routeD))

# Violin Plot

ggplot(df,

aes(x=Route,

y=Delivery_Time,
```

```
fill=Route))+

geom_violin(trim=FALSE)+

geom_boxplot(width=0.1,

fill="white")+

labs(title="Violin Plot of Delivery Time by Route",

x="Route",

y="Delivery Time (Minutes)")+

theme_minimal()
```

Interpretation (Violin Plot)

- **Route B** has the lowest median delivery time, making it the fastest route.
- **Route C** has the widest distribution, indicating the greatest variability in delivery times.
- **Route A** and **Route D** have moderate delivery times with less variation than Route C.

Task 12 – Ridgeline Plot



Expected Output

Violin Plot

- Four violin plots representing Route A, Route B, Route C, and Route D.
- Each violin contains a boxplot showing the median and quartiles.

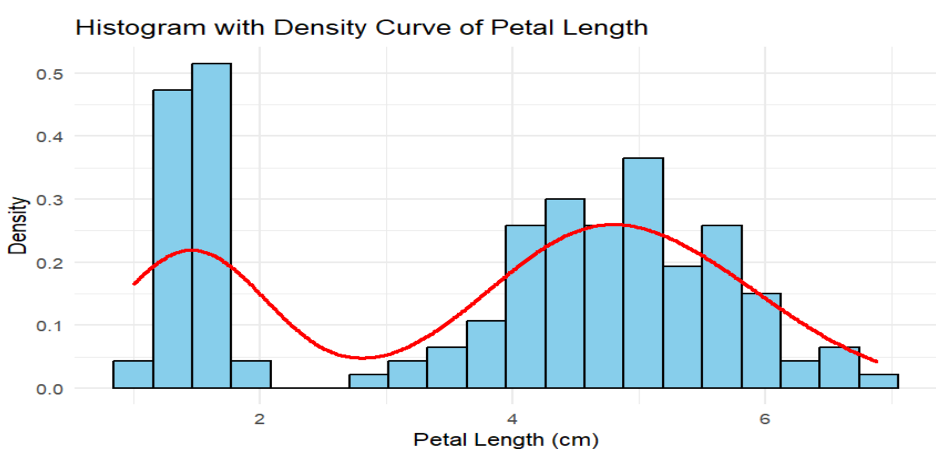
Ridgeline Plot

- Four overlapping density curves.
- Routes arranged from fastest (lowest mean delivery time) at the bottom to slowest (highest mean delivery time) at the top.

Overall Interpretation

- Route B has the shortest average delivery time, making it the most efficient route.
- Route C has the highest average delivery time and the greatest spread, indicating more variability and inconsistency.
- Route A and Route D show moderate delivery times with relatively balanced distributions.
- The violin plot highlights the spread and median of each route, while the ridgeline plot makes it easy to compare the distribution shapes across all routes.

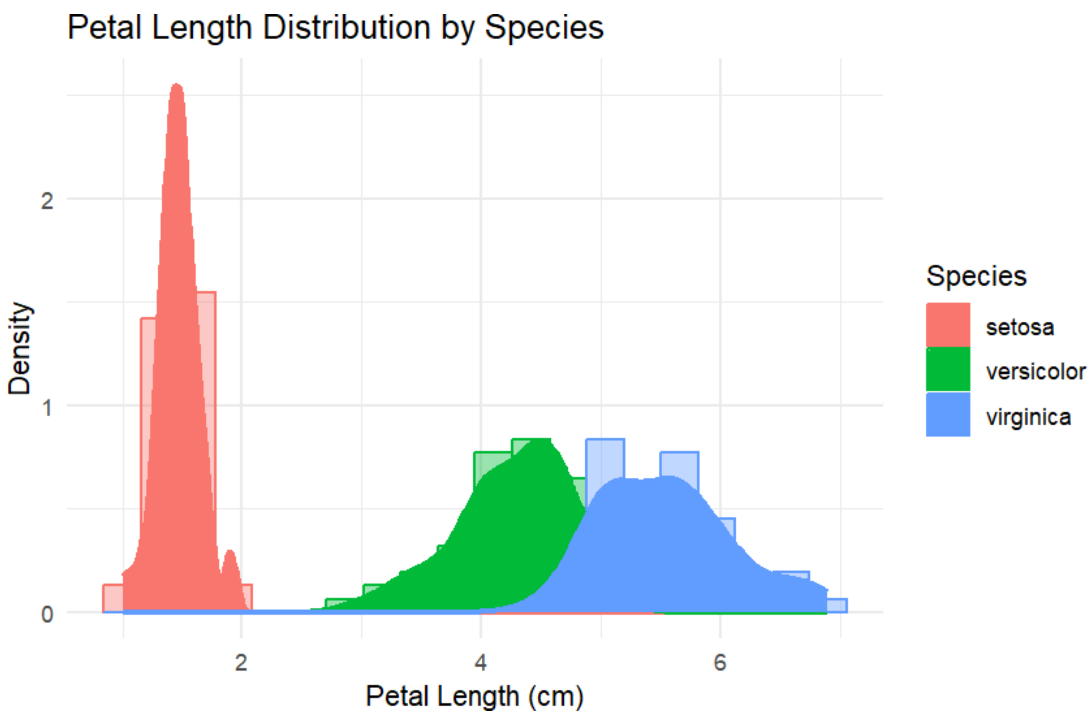
QUESTION-5



CODE:

```
# Load library

library(ggplot2)

# Histogram with Density Curve

ggplot(iris,

       aes(x = Petal.Length)) +

  geom_histogram(

    aes(y = after_stat(density)),

    bins = 20,

    fill = "skyblue",

    color = "black"

  ) +

  geom_density(

    color = "red",

    linewidth = 1

  ) +

  labs(

    title = "Histogram with Density Curve of Petal Length",

    x = "Petal Length (cm)",

    y = "Density"

  ) +

  theme_minimal()
```

Task 14 – Histogram & KDE Split by Species



- **Expected Output**

1. Overall Histogram with KDE

A histogram displaying the distribution of **Petal Length** using **20 bins**.

A smooth **Kernel Density Estimation (KDE)** curve is overlaid to illustrate the overall distribution pattern.

2. Histogram with KDE by Species

Three overlapping histograms are displayed for:

Setosa

Versicolor

Virginica

Each species is represented by a different color along with its corresponding KDE curve, making comparison of their petal length distributions easier.

- **Interpretation**

Overall Histogram

Petal length values are distributed approximately between **1 cm and 7 cm**.

The KDE curve provides a smooth representation of the overall data distribution.

Since the dataset contains measurements from three different species, the distribution is not perfectly symmetrical and may show multiple peaks.

- **Histogram by Species**

Setosa has the shortest petal lengths and forms a well-separated distribution.

Versicolor shows medium petal lengths with a moderate level of variation.

Virginica has the longest petal lengths and the widest spread among the three species.